

UNIT IV

Synchronous & Asynchronous Sequential Circuits: Analysis of clocked sequential circuits with state machine designing, State reduction and assignments, Design procedure. Analysis procedure of Asynchronous sequential circuits, circuit with latches, Design procedure, Reduction of state and flow table, Race-free state assignment, Hazards.

PART 1 – Synchronous Sequential Circuits

1 What is a Synchronous Sequential Circuit?

A synchronous circuit:

- Uses clock signal
- All flip-flops change state only on clock edge
- More stable and reliable

Example:

CPU registers

Counters

State machines

2 State Machine Concept

A sequential circuit can be represented using:

- States
- Inputs
- Outputs
- Transitions

This representation is called a **State Machine**.

Two types:

Type	Output Depends On
Moore Machine	Present state only
Mealy Machine	Present state + input

Moore vs Mealy (Very Important)

Feature	Moore	Mealy
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Output Depends On	State only	State + Input
Output Change	Only at clock	Can change immediately
Number of States	More	Less
Stable	More stable	Faster response

3 Analysis of Clocked Sequential Circuits

Suppose you are given:

- Flip-flops
- Logic gates
- Circuit diagram

You must find:

- State table
- State diagram
- Output behavior

Analysis Procedure (Step-by-Step)

1. Identify flip-flop type (D, JK, T)
2. Write characteristic equations
3. Find next state equation
4. Create state table
5. Draw state diagram

Example (Conceptual)

If D flip-flop is used:

$$Q^+ = D$$

So next state directly equals D input.

4 Design of Synchronous Sequential Circuits

If question gives:

"Design a sequence detector"

or

"Design a counter"

Follow this procedure:

Design Procedure (Step-by-Step)

1. Understand problem
2. Draw state diagram
3. Write state table
4. Reduce states (if possible)
5. Assign binary codes to states
6. Choose flip-flop type
7. Find excitation table
8. Derive flip-flop input equations
9. Draw final circuit

5 State Reduction

Sometimes state diagram has extra states.

We remove equivalent states to simplify circuit.

Two states are equivalent if:

- They have same output
- They go to same next states

This reduces hardware.

6 State Assignment

After reduction:

Assign binary codes to states.

Example:

State A → 00

State B → 01

State C → 10

State D → 11

Goal:

Avoid unnecessary complexity.

PART 2 – Asynchronous Sequential Circuits

1 What is Asynchronous Sequential Circuit?

- No clock signal
- Output changes immediately when input changes
- Uses latches instead of flip-flops

Less stable than synchronous circuits.

Where Used?

- Small control circuits
- Simple switching systems

2 Analysis Procedure of Asynchronous Circuits

Given:

- Latches
- Combinational logic

Steps:

1. Write Boolean equations
2. Construct flow table
3. Determine stable states
4. Draw state diagram

3 Flow Table

Flow table is like state table but:

- Shows stable states
- Shows unstable transitions

Used only in asynchronous circuits.

4 Circuit with Latches

Asynchronous circuits use:

- SR latch
- D latch

Since no clock, timing issues can occur.

5 Reduction of State and Flow Table

Similar to synchronous state reduction.

Remove equivalent rows.

This simplifies design.

6 Race Condition

Race occurs when:

Two state variables change at same time.

Because of delay difference:

Wrong state may occur.

Types of Race

1. Critical race → Wrong output
2. Non-critical race → Correct output eventually

7 Race-Free State Assignment

To avoid race:

Assign binary codes so that:

Only one bit changes during transition.

Example:

Use Gray code style assignment.

This prevents multiple flip-flop changes at same time.

8 Hazards

Hazards are unwanted temporary changes in output.

Because of:

Propagation delay in logic gates.

Types of Hazards

1. Static Hazard

Output should remain 1 but becomes 0 temporarily.

2. Dynamic Hazard

Output changes multiple times before stabilizing.

3. Essential Hazard

Occurs in asynchronous circuits due to unequal delays.

How to Remove Hazards?

- Add redundant logic terms
- Use proper state assignment
- Balance propagation delay

① Synchronous vs Asynchronous Sequential Circuit ★ (Very Important)

Feature	Synchronous Circuit	Asynchronous Circuit
Clock Signal	Required	Not required
State Change	Only on clock edge	Immediately when input changes
Stability	More stable	Less stable
Design Complexity	Easier	More complex
Speed	Controlled by clock	Faster response
Hazard Risk	Less	More
Used In	CPU, processors	Small control circuits

Exam tip: Always write “**clock controlled**” vs “**no clock**”.

② Moore Machine vs Mealy Machine ★

Feature	Moore Machine	Mealy Machine
Output Depends On	Present state only	Present state + input
Output Change	Only at clock edge	Can change immediately
Number of States	More	Less
Stability	More stable	Faster response
Output Timing	Delayed by one clock	Immediate

Important line:

Moore → Output = f(state)

Mealy → Output = f(state, input)

③ State Table vs Flow Table

Feature	State Table	Flow Table
Used In	Synchronous circuits	Asynchronous circuits
Based On	Clocked transitions	Input changes
Shows	Present state & next state	Stable & unstable states
Complexity	Simpler	More detailed

4 Latch-Based vs Flip-Flop-Based Circuits

Feature	Latch Circuit	Flip-Flop Circuit
Trigger Type	Level triggered	Edge triggered
Used In	Asynchronous circuits	Synchronous circuits
Stability	Less stable	More stable
Clock	No clock required	Clock required

5 Critical Race vs Non-Critical Race

Feature	Critical Race	Non-Critical Race
Final State	Wrong state	Correct state
Problem Severity	Serious	Temporary
Must Avoid	Yes	Not necessary
Occurs In	Asynchronous circuits	Asynchronous circuits

6 Static Hazard vs Dynamic Hazard

Feature	Static Hazard	Dynamic Hazard
Output Behavior	Output should remain same but glitches	Output changes multiple times
Occurs Due To	Unequal gate delays	Multiple path delays
Type	Static-1 or Static-0	Only dynamic
Common In	Combinational circuits	Complex circuits

7 State Reduction vs State Assignment

Feature	State Reduction	State Assignment
Purpose	Reduce number of states	Assign binary codes to states
When Done	Before assignment	After reduction
Goal	Minimize hardware	Avoid race conditions

8 Synchronous Design Procedure vs Asynchronous Design Procedure

Feature	Synchronous Design	Asynchronous Design
First Step	Draw state diagram	Write flow table

Memory Element	Flip-flops	Latches
Clock Required	Yes	No
Race Problem	Rare	Common
Design Difficulty	Moderate	Difficult

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